

Discussion section 1

Huangchen Zhou

Jan 13, 2026

1 Chp 1, Q7

A rat is trapped in a maze with three doors and some hidden cheese. If the rat takes door one, he will wander around the maze for 2 minutes and return to where he started. If he takes door two, he will wander around the maze for 3 minutes and return to where he started. If he takes door three, he will find the cheese after 1 minute. If the rat returns to where he started he immediately picks a door to pass through. The rat picks each door uniformly at random. How long, on average, will the rat wander before finding the cheese?

Proof. (Compare Example 1.21) Let X be the time until the rat finds the cheese. Then

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}[X|\text{first door is door one}] \cdot \mathbb{P}(\text{first door is door one}) \\ &\quad + \mathbb{E}[X|\text{first door is door two}] \cdot \mathbb{P}(\text{first door is door two}) \\ &\quad + \mathbb{E}[X|\text{first door is door three}] \cdot \mathbb{P}(\text{first door is door three}) \\ &= (\mathbb{E}[X] + 2) \cdot \frac{1}{3} + (\mathbb{E}[X] + 3) \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}\end{aligned}$$

Thus $\mathbb{E}[X] = 6$. □

2 Chp 1, Q10

A die is rolled until a 3 occurs. By conditioning on the outcome of the first roll, find the probability that an even number of rolls is needed.

Proof. (Compare Example 1.21) Let X be the number of trials to get the

first 3, then

$$\begin{aligned}
 \mathbb{P}(X \text{ is even}) &= \mathbb{P}(X \text{ is even} | \text{first trial is 3}) \cdot \mathbb{P}(\text{first trial is 3}) \\
 &\quad + \mathbb{P}(X \text{ is even} | \text{first trial is not 3}) \cdot \mathbb{P}(\text{first trial is not 3}) \\
 &= 0 + \mathbb{P}(X \text{ is odd}) \cdot \frac{5}{6} \\
 &= 0 + (1 - \mathbb{P}(X \text{ is even})) \cdot \frac{5}{6}
 \end{aligned}$$

Thus $\mathbb{P}(X \text{ is even}) = 5/11$. □

3 Chp 1, Q15

Let X and Y be uniformly distributed on the disk of radius 1 centered at the origin. Find the conditional distribution of Y given $X = x$.

Proof. (Compare Example 1.21)

By definition, the joint pmf is $f(x, y) = \frac{1}{\pi}$, if $x^2 + y^2 \leq 1$ and $f(x, y) = 0$ otherwise.

And by integrating against y , the marginal distribution of x is given by $f_X(x) = \frac{2}{\pi}\sqrt{1-x^2}$ if $-1 \leq x \leq 1$ and $f_X(x) = 0$ otherwise. Thus the conditional density function is given by

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \begin{cases} \frac{1}{2\sqrt{1-x^2}}, & \text{if } -\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}; \\ 0 & \text{otherwise.} \end{cases}$$

Thus the conditional distribution of Y given $X = x$ is uniform on the interval $[-\sqrt{1-x^2}, \sqrt{1-x^2}]$. □

4 Chp 1, Q17

Let X be a Poisson random variable with $\lambda = 3$. Find $\mathbb{E}[X|X > 2]$.

Proof. (Compare Example 1.23)

$$\begin{aligned}
 \mathbb{E}[X|X > 2] &= \sum_{k=0}^{\infty} k \cdot \mathbb{P}(X = k | X > 2) = \sum_{k=0}^{\infty} k \cdot \frac{\mathbb{P}(X = k \text{ and } X > 2)}{\mathbb{P}(X > 2)} \\
 &= \sum_{k=3}^{\infty} k \cdot \frac{\mathbb{P}(X = k)}{\mathbb{P}(X > 2)} = \frac{\mathbb{E}[X] - 0 \cdot \mathbb{P}(X = 0) - 1 \cdot \mathbb{P}(X = 1) - 2 \cdot \mathbb{P}(X = 2)}{1 - \mathbb{P}(X = 0) - \mathbb{P}(X = 1) - \mathbb{P}(X = 2)}
 \end{aligned}$$

Since $\mathbb{P}(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$ and $\mathbb{E}[X] = \lambda$ for Poisson distribution, we have

$$\mathbb{E}[X|X > 2] = \frac{3 - 3e^{-3} - 2 \cdot 4.5e^{-3}}{1 - e^{-3} - 3e^{-3} - 4.5e^{-3}} \approx 4.165246$$

□

5 Chp 2, Q6

A tetrahedron die has four faces labeled 1,2,3, and 4. In repeated independent rolls of the die $R_0, R_1, \dots$, let $X_n = \max\{R_0, \dots, R_n\}$ be the maximum value after $n + 1$ rolls, for $n \geq 0$.

(a) Give an intuitive argument for why $X_0, X_1, \dots$ is a Markov chain, and exhibit the transition matrix.

Proof. It suffices to check the memoryless property, namely

$$\mathbb{P}(X_{n+1} = x_{n+1} | X_n = x_n \cdots, X_1 = x_1, X_0 = x_0) = \mathbb{P}(X_{n+1} = x_{n+1} | X_n = x_n),$$

for any n . Intuitively, the memoryless property holds since $X_n = \max\{X_{n-1}, R_n\}$, i.e. to know the maximum value after $n + 1$ rolls, it is equivalent to looking at the maximum value X_{n-1} after n rolls, and comparing it with the $n + 1$ roll R_n .

By homogeneity, $P_{ij} = \mathbb{P}(X_1 = j | X_0 = i)$. The transition matrix is given by:

$$\begin{bmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 1/2 & 1/4 & 1/4 \\ 0 & 0 & 3/4 & 1/4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Sanity check: rows add up to one! □

(b) Find $\mathbb{P}(X_3 \geq 3)$.

Proof. Among the first four rolls, at least one of them is 3 or 4. By taking the complement of the event, $\mathbb{P}(X_3 \geq 3) = 1 - (2/4)^4 = 15/16$. □